

典型分布式（云）技术及系统

分布式计算系统课程实验

典型分布式（云）技术及系统

- Google云范例（Hadoop）
 - GFS
 - Chubby
 - Bigtable
 - MapReduce
- Spark
- OpenStack
- Storm/Heron
- Scribe（Facebook）
- Kafka（LinkedIn）
- Flume（Cloudera）
- 分布式训练与推理

Hadoop

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Apache Hadoop

The Apache™ Hadoop® project develops open-source software for reliable, scalable, distributed computing.

The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. Rather than rely on hardware to deliver high-availability, the library itself is designed to detect and handle failures at the application layer, so delivering a highly-available service on top of a cluster of computers, each of which may be prone to failures.

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Latest news

[Ozone 0.3.0-alpha is released](#)

2018 Nov 22

Modules

The project includes these modules:

Related projects

Other Hadoop-related projects at Apache include:

Spark



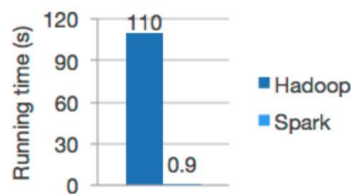
Download Libraries Documentation Examples Community Developers Apache Software Foundation

Apache Spark™ is a unified analytics engine for large-scale data processing.

Speed

Run workloads 100x faster.

Apache Spark achieves high performance for both batch and streaming data, using a state-of-the-art DAG scheduler, a query optimizer, and a physical execution engine.



Logistic regression in Hadoop and Spark

Latest News

- Spark+AI Summit (April 23-25th, 2018, San Francisco) agenda posted (Dec 19, 2018)
- Spark 2.4.0 released (Nov 02, 2018)
- Spark 2.3.2 released (Sep 24, 2018)
- Spark+AI Summit (October 2-4th, 2018, London) agenda posted (Jul 24, 2018)

[Archive](#)



<http://spark.apache.org/downloads.html>

Ease of Use

```
df = spark.read.json("logs.json") df.where
```

Download Spark

openstack

This is the current supported release.

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Welcome to OpenStack Documentation

What is OpenStack? OpenStack is a cloud operating system that controls large pools of compute, storage, and networking resources throughout a datacenter, all managed through a dashboard that gives administrators control while empowering their users to provision resources through a self-service portal.

✓ 截图时隐藏当前窗口

What Are The Next Steps?

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Get OpenStack ➡

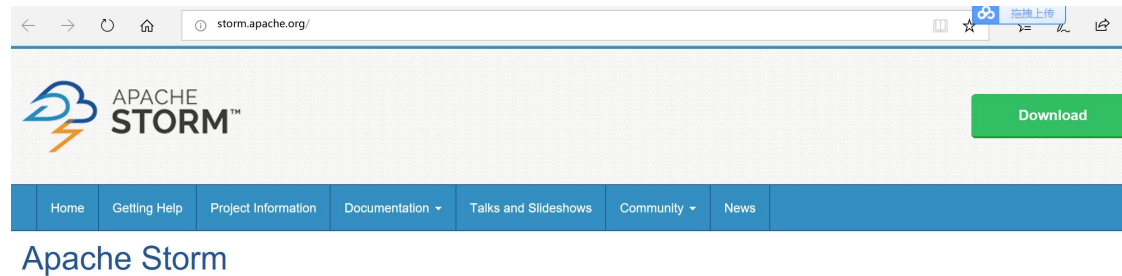
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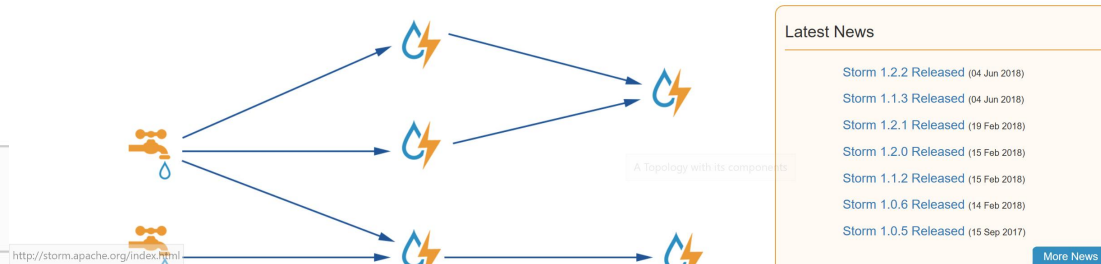


Narayan Desai

Storm/Heron



Why use Storm?

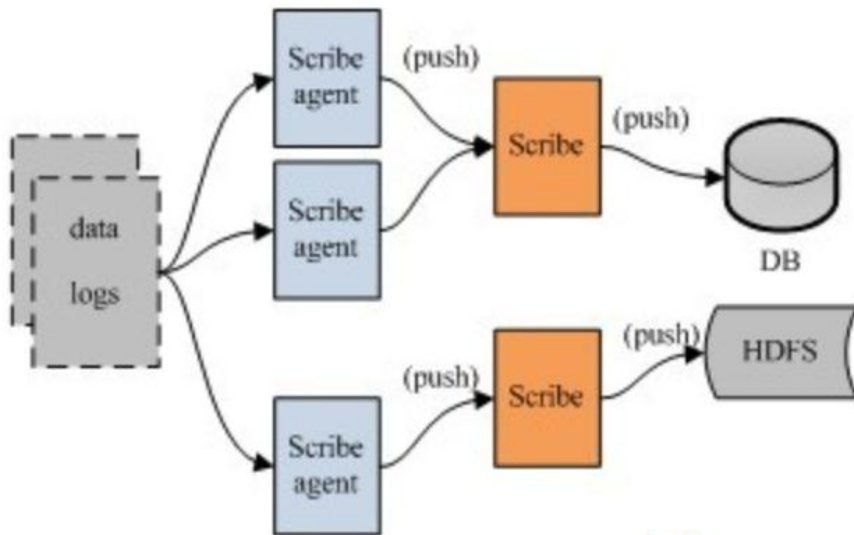


Apache Storm is a free and open source distributed realtime computation system. Storm makes it easy to reliably process unbounded streams of data, doing for realtime processing what Hadoop did for batch processing. Storm is simple, can be used with any programming language, and is a lot of fun to use!

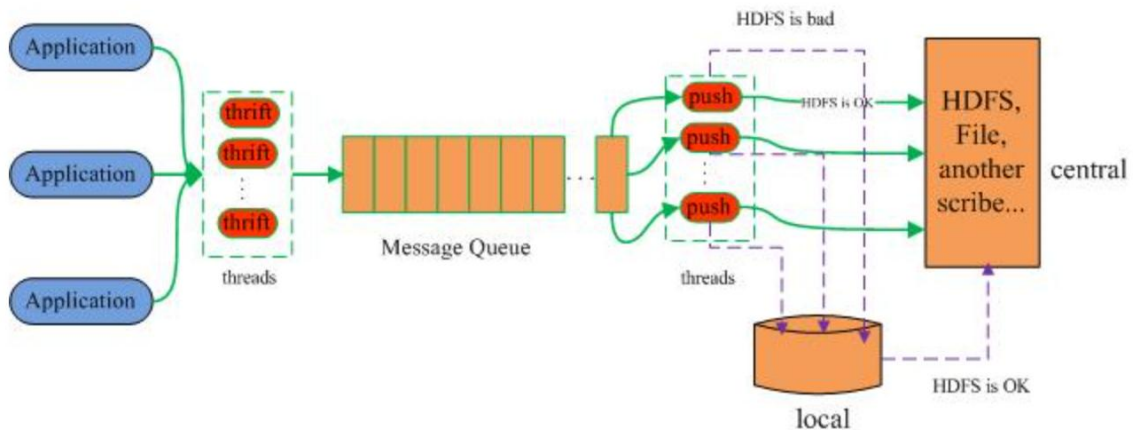
Storm has many use cases: realtime analytics, online machine learning, continuous computation, distributed RPC, ETL, and more. Storm is fast: a benchmark clocked it at over **a million tuples processed per second per node**. It is scalable, fault-tolerant, guarantees your data will be processed, and is easy to set up and operate.

Storm integrates with the queueing and database technologies you already use. A Storm topology consumes streams of data and processes those streams in arbitrarily complex ways, repartitioning the streams between each stage of the computation however needed. Read more in the tutorial.

Scribe (Facebook)



架构



Kafka (LinkedIn)



HOME

INTRODUCTION

QUICKSTART

USE CASES

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<http://kafka.apache.org/documentation/#producerapi>

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PUBLISH & SUBSCRIBE

Read and write streams of data like a messaging system.

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PROCESS

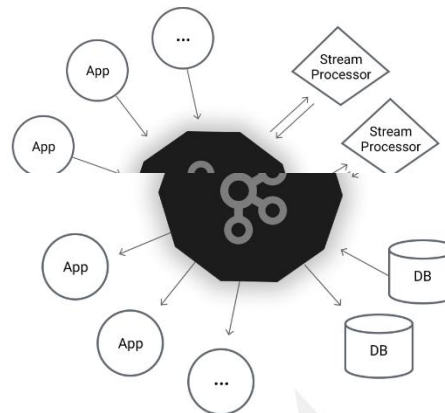
Write scalable stream processing applications that react to events in real-time.

[Learn more »](#)

STORE

Store streams of data safely in a distributed, replicated, fault-tolerant cluster.

[Learn more »](#)



Kafka® is used for building real-time data pipelines and streaming apps. It is horizontally scalable, fault-tolerant, wicked fast, and runs in production in thousands of companies.

[Learn More](#)

Flume (Cloudera)

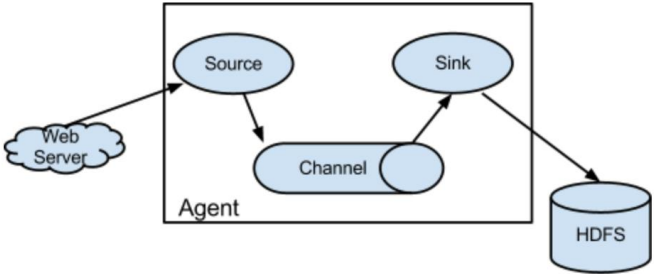


Apache Flume



Welcome to Apache Flume

Flume is a distributed, reliable, and available service for efficiently collecting, aggregating, and moving large amounts of log data. It has a simple and flexible architecture based on streaming data flows. It is robust and fault tolerant with tunable reliability mechanisms and many failover and recovery mechanisms. It uses a simple extensible data model that allows for online analytic application.



```
graph LR; WS[Web Server] --> S((Source)); S --> C((Channel)); C --> K((Sink)); K --> HDFS[(HDFS)]; subgraph Agent; S; C; K; end
```

Apache Flume

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Resources

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Apache

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